COMSATS University Islamabad

Theory Assignment # 4

Course Name: Applications of ICT

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Class/ Section: BDS-1A

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For the given problems implement the designed solutions in python.

Question 1: Calculate the age by taking date of birth from the user as input and display age as follows: Your age is XX years, XX months, and XX days.

Code:

```python
# Take current date
current_day = int(input("Enter current day: "))
current_month = int(input("Enter current month: "))
current_year = int(input("Enter current year: "))

# Take date of birth
birth_day = int(input("Enter birth day: "))
birth_month = int(input("Enter birth month: "))
birth_year = int(input("Enter birth year: "))

# Days in each month
def days_in_month(month, year):
    # February check (simple, ignoring leap years)
    if month == 2:
        return 28
    # Months with 30 days
    elif month == 4 or month == 6 or month == 9 or month == 11:
        return 30
    # Rest have 31 days
    else:
        return 31

```

```python
# Calculate days
if current_day < birth_day:
    # Borrow days from previous month
    current_day += days_in_month(current_month - 1 if current_month > 1 else 12, current_year)
    current_month -= 1

age_days = current_day - birth_day

# Calculate months
if current_month < birth_month:
    current_month += 12
    current_year -= 1

age_months = current_month - birth_month

# Calculate years
age_years = current_year - birth_year

# Output
print("Your age is", age_years, "years,", age_months, "months, and", age_days, "days.")
```

Output:

```
Enter current day: 12
Enter current month: 2
Enter current year: 2025
Enter birth day: 14
Enter birth month: 4
Enter birth year: 2007
Your age is 17 years, 9 months, and 29 days.
```

Question 2: For any integer received from the user between 0 and 1000, adds all the digits in the integer. For example, if an integer is 932, the sum of all its digits is 14.

Code:

```python
num = int(input("Enter an integer between 0 and 1000: "))

# Initialize sum
sum_of_digits = 0

# Loop until num becomes 0
while num > 0:
    digit = num % 10      # Get last digit
    sum_of_digits += digit # Add it to sum
    num = num // 10        # Remove last digit

print("The sum of the digits is:", sum_of_digits)
```

Output:

```
Enter an integer between 0 and 1000: 66
The sum of the digits is: 12
```

Question 3: How to determine any number to be even or Odd ?

Code:

```
num = int(input("Enter a number: "))

if num % 2 == 0:
    print(num, "is Even")
else:
    print(num, "is Odd")

```

Output:

```
Enter a number: 5
5 is Odd
```

Question 4: Design the working of calculator by the use of functions.

Code:

```python
print("Simple Calculator")
# Function for addition
def add(a, b):
    return a + b

# Function for subtraction
def subtract(a, b):
    return a - b

# Function for multiplication
def multiply(a, b):
    return a * b

# Function for division
def divide(a, b):
    return a / b

while True:
    print("\nChoose an operation:")
    print("1. Add")
    print("2. Subtract")
    print("3. Multiply")
    print("4. Divide")
    print("5. Exit")
```

```python
    choice = input("Enter your choice (1-5): ")

    if choice == "5":
        print("Exiting...")
        break

    num1 = float(input("Enter first number: "))
    num2 = float(input("Enter second number: "))

    if choice == "1":
        print("Result:", add(num1, num2))
    elif choice == "2":
        print("Result:", subtract(num1, num2))
    elif choice == "3":
        print("Result:", multiply(num1, num2))
    elif choice == "4":
        print("Result:", divide(num1, num2))
    else:
        print("Invalid choice. Try again!")
```

Output:

<u>If we want to ADD.</u>

```
Choose an operation:
1. Add
2. Subtract
3. Multiply
4. Divide
5. Exit
Enter your choice (1-5): 1
Enter first number: 2
Enter second number: 2
Result: 4.0
```

<u>If we want to SUBSTRACT.</u>

```
Choose an operation:
1. Add
2. Subtract
3. Multiply
4. Divide
5. Exit
Enter your choice (1-5): 2
Enter first number: 3
Enter second number: 2
Result: 1.0
```

If we want to MULTIPLY.

```
Choose an operation:
1. Add
2. Subtract
3. Multiply
4. Divide
5. Exit
Enter your choice (1-5): 3
Enter first number: 10
Enter second number: 5
Result: 50.0
```

If we want to DIVIDE.

```
Choose an operation:
1. Add
2. Subtract
3. Multiply
4. Divide
5. Exit
Enter your choice (1-5): 4
Enter first number: 10
Enter second number: 5
Result: 2.0
```